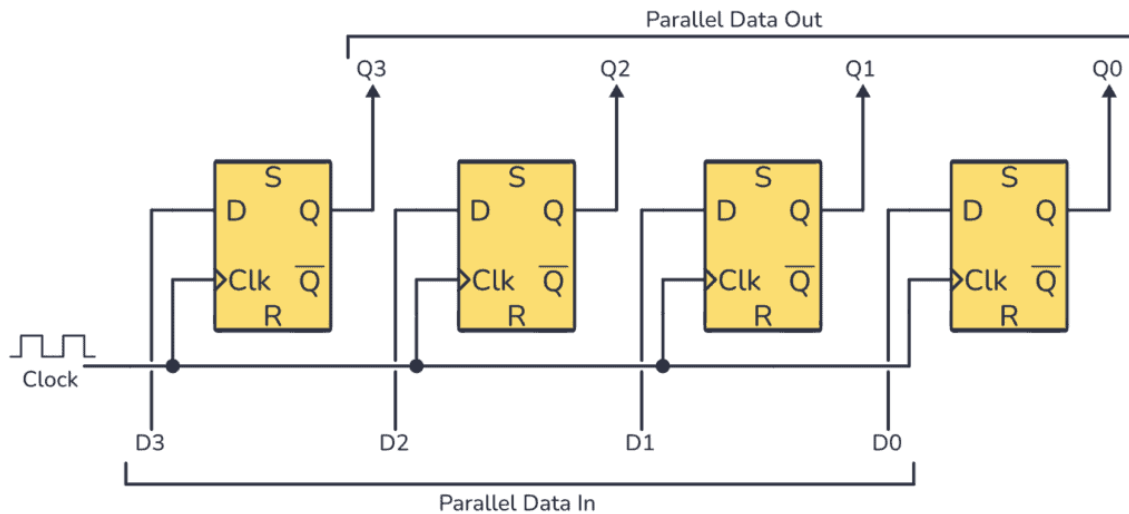


Day 32

4-bit Parallel-In Parallel-Out (PIPO) Register in Verilog

Circuit Diagram :



Verilog Code

verilog code:

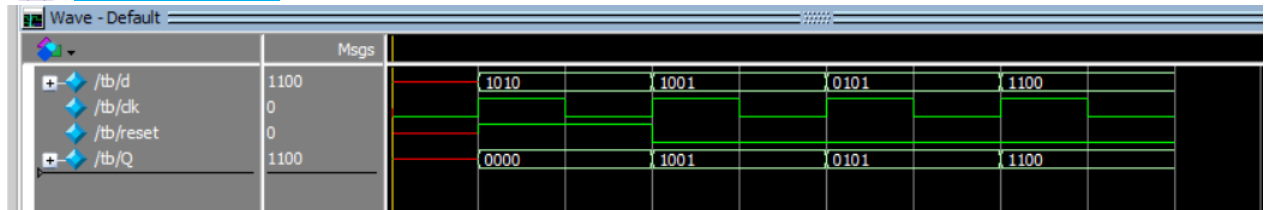
```
1  module PIPO(clk,reset,d,Q);
2  input [3:0]d;
3  input clk,reset;
4  output reg [3:0]Q;
5  always @(posedge clk)
6  if(reset) Q<=4'b0000;
7  else
8  begin
9  Q[3]<=d[3];
10 Q[2]<=d[2];
11 Q[1]<=d[1];
12 Q[0]<=d[0];
13 //short we can also use Q<=d
14 end
15 endmodule
16
17
```

Testbench Code

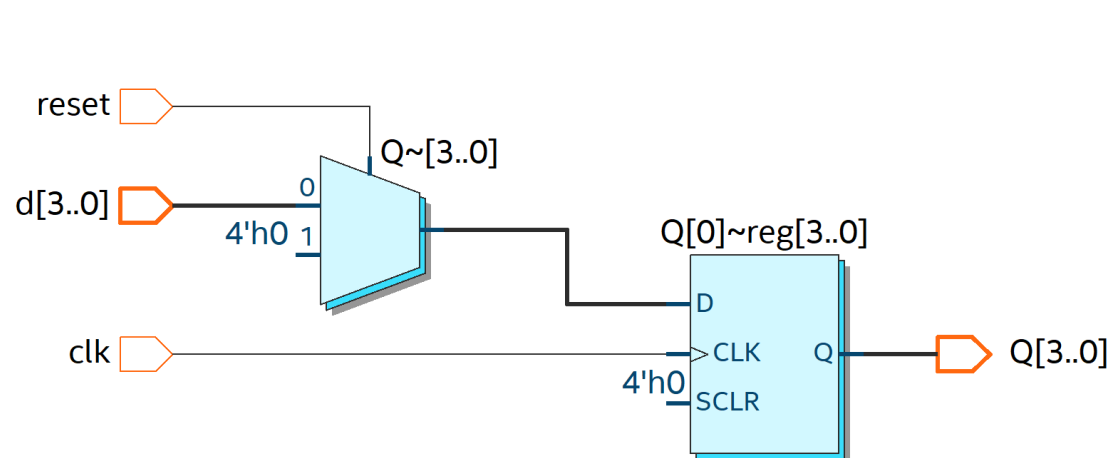
Tb code :

```
1 module tb;
2   reg [3:0]d;
3   reg clk,reset;
4   wire [3:0]Q;
5   PIPO dut(clk,reset,d,Q);
6   initial
7   begin
8     clk=0;
9     forever #5 clk=~clk;
10  end
11  initial begin
12    #5
13    reset=1; d=4'b1010; #10
14    reset=0; d=4'b1001; #10
15    d=4'b0101; #10
16    d=4'b1100; #10
17    $finish;
18  end
19 endmodule
```

Waveform



Schematic





EDA Tools Used

- IntelQuartusPrime
- ModelSim

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